

NEXAM SYSTEMS

UNCREWED AIRCRAFT SYSTEMS · SYSTEMS ENGINEERING

ARCHITECTURE DESCRIPTION

Multi-Vehicle Orchestration Subsystem

STRATUM Command & Control · Dagger Air Vehicle

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System of interest	Multi-vehicle orchestration subsystem, STRATUM-Desktop
Ground baseline	STRATUM-Desktop @ 4095ba3
Air baseline	PX4-Autopilot-NX (Dagger), working tree
Status	For internal review — decisions D1-D9 proposed, five rulings open

COMPANY PROPRIETARY

This document describes an architecture as a set of decisions and their consequences. It is derived from a direct scan of both code baselines named above; every claim about existing behaviour carries a file and line reference.

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1 Purpose, scope and definitions

An architecture is a set of decisions. Everything else is consequence. This document records nine decisions for coordinating three Dagger air vehicles onto a standoff ring, the evidence from both code baselines that forced them, and the five rulings still outstanding.

1.1 Purpose

This document defines the architecture of a multi-vehicle orchestration subsystem residing inside STRATUM-Desktop. The subsystem coordinates up to three Dagger air vehicles onto distinct stations on a standoff ring about a single operator-designated target. It is written to be reviewed, argued with, and then built from; §11 lists the questions that must be answered before construction begins.

1.2 Scope

In scope: target designation, ring definition, slot assignment, launch and transit sequencing, arrival detection, separation assurance, and degraded-mode behaviour for a three-vehicle ensemble. Out of scope, deliberately and explicitly: multiple simultaneous targets; fixed-wing standoff geometry; inter-vehicle datalinks; and — pending the ruling in §11.5 — any transition from an orchestrated standoff into a terminal engagement.

1.3 Definitions

Terms are fixed here and used with these meanings throughout. A term defined here is never used loosely elsewhere in the document.

Term	Definition
Target	A single geodetic point designated by the operator. One per mission. Primitive.
Standoff ring	The horizontal circle of radius R centred on the target at height H above home. Derived , never stored independently — it therefore cannot disagree with the target.
Ring slot	A station on the ring, identified by bearing θ from target to hold point ($0 = \text{North}$, clockwise). One vehicle per slot; one slot per vehicle.
Transit level	The altitude at which a vehicle performs its horizontal run-in. Distinct from the slot's hold height, and the basis of vertical deconfliction.
Agent	STRATUM's per-vehicle orchestration object: one <code>Vehicle*</code> , one slot, one state machine. An agent pins its vehicle for life and never reads <code>activeVehicle</code> .
Commit	The act of sending a vehicle its standoff geometry and switching it into the Standoff flight mode. The irreversible step in the sequence.
Separation bubble	The keepout sphere carried by each vehicle. The nominal 1 m figure is retained as a hard body keepout but is not the operative criterion; §7.5 derives what is.
Deconfliction by construction	Separation guaranteed by geometry and schedule chosen before flight, rather than by a runtime avoidance controller. The organising principle of §7.
Tradespace	The set of feasible architectures spanned by the decisions in §3, together with the consequences each choice imposes.

1.4 System context

The system of interest sits between one operator and three air vehicles. It has no peers, no external interfaces beyond the existing MAVLink transport, and no authority that the operator has not granted. Figure 1 fixes the boundary.

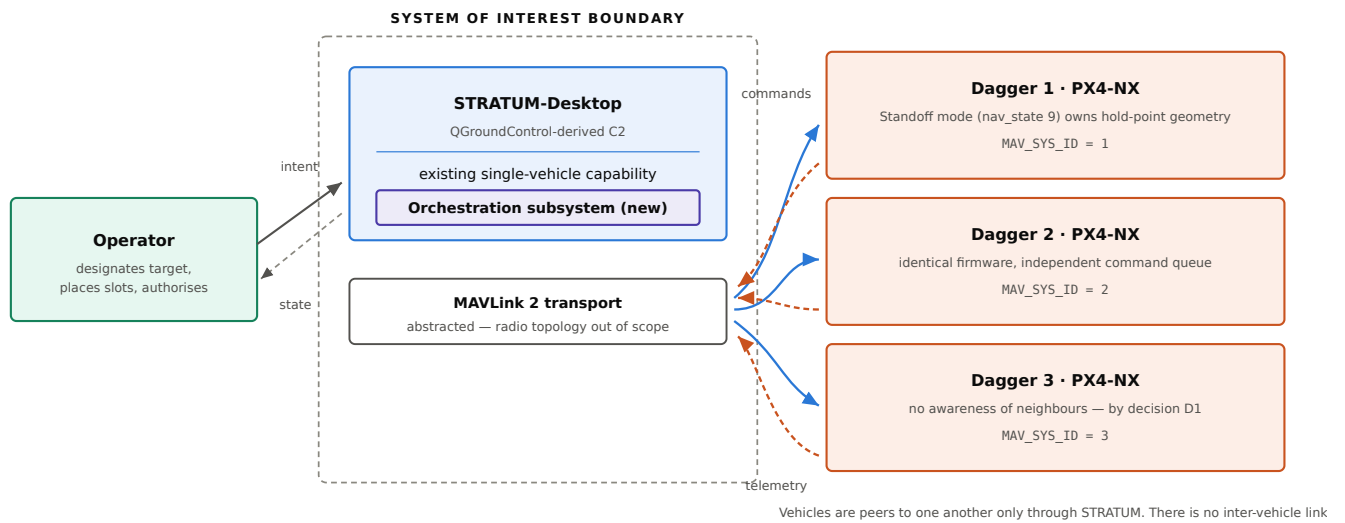


Figure 1 — System context. The orchestration subsystem is additive: it sits alongside the existing single-vehicle capability inside STRATUM and reaches the vehicles through the same per-vehicle command path. Every coordinating decision is made on the ground; no vehicle is aware that the others exist.

2 The as-built baseline

An architecture built on an assumed baseline is a hypothesis. This section records what the two code baselines actually do, with references, because four of the nine decisions in §3 were changed by what the scan found.

2.1 The single-vehicle control chain

The essential division of labour in the existing system is already correct, and it is not the one an outside reader would assume. STRATUM does *not* compute the standoff hold point. It transmits the **target** plus three scalars; the firmware translates the target, resolves the hold altitude against home, and computes the facing yaw itself.

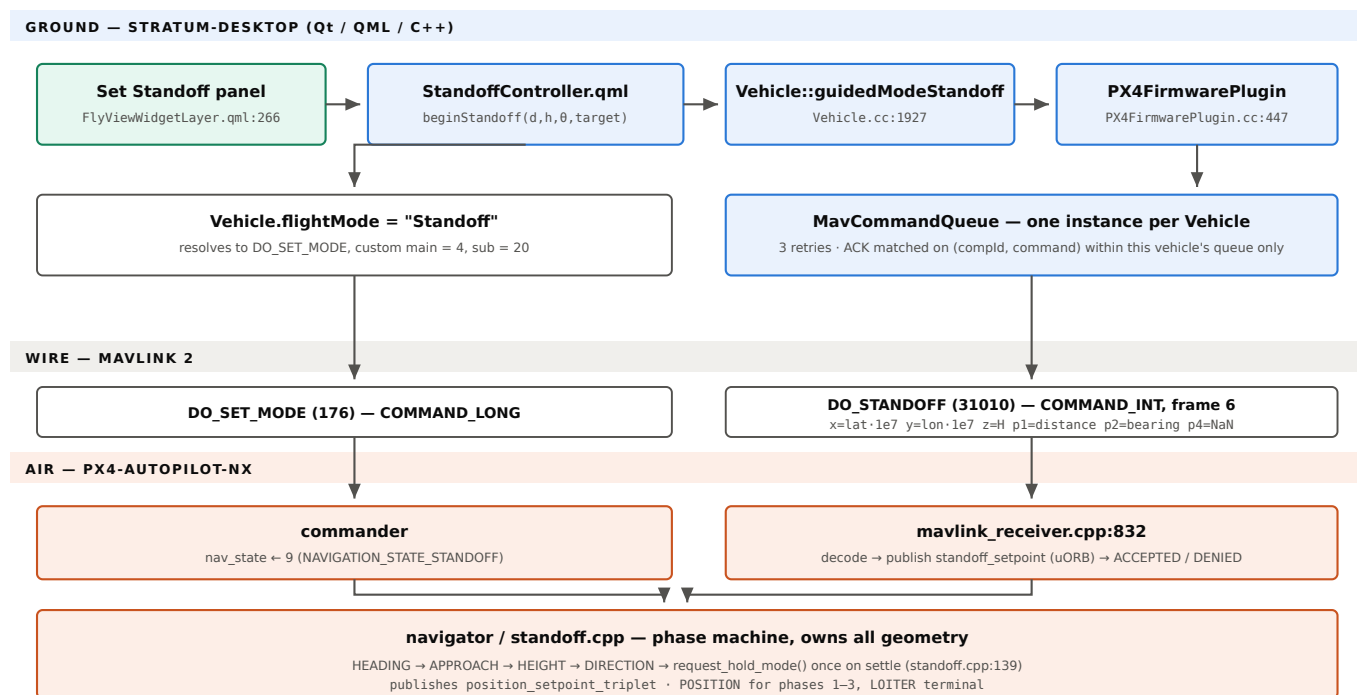
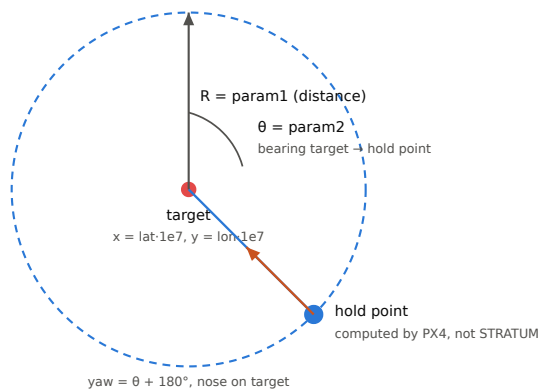


Figure 2 — As-built single-vehicle control chain. Two independent messages carry the mission: the geometry (31010) and the mode selection (176). PX4 tolerates either order; STRATUM sends geometry first so the mode never activates against an empty setpoint. Everything below `StandoffController.qml` is already scoped to one `Vehicle` instance — including the command queue, its retries and its ACK matching.

2.2 The firmware's geometry and staging

The Standoff mode decomposes the manoeuvre into four ordered phases rather than commanding one diagonal move. The rationale in the firmware documentation is sensor settling: a payload that must acquire the target benefits from turning, translating and changing height as discrete, observable steps. That decision, taken for a single vehicle and for a wholly different reason, is what makes vertical deconfliction free for three vehicles — see finding F3 and §7.3.

PLAN VIEW — GEOMETRY OWNED BY PX4



STDF_SEQ ∈ {2,4} folds HEIGHT into APPROACH

— which collapses the vertical separation §7.3 depends on. Deconfliction-critical parameter.

PHASE MACHINE — navigator/standoff.cpp

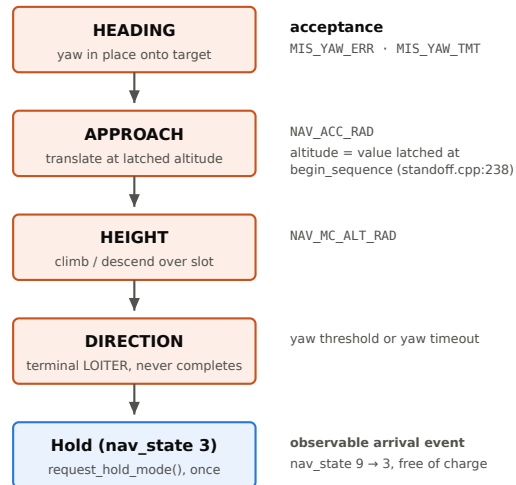


Figure 3 — Standoff geometry and the PX4 phase machine. The firmware translates the target by param1 along param2, resolves hold altitude as home + z, and yaws to face the target. The APPROACH phase flies at the altitude latched when the sequence begins — the hook on which the entire deconfliction architecture rests.

2.3 What is already multi-vehicle

Everything below the QML layer. This is the most consequential finding of the scan, because it converts the work from a refactor into an addition — and a refactor of a working weapon-system C2 application is a risk that does not need to be taken.

Element	State
MultiVehicleManager	Stock and unmodified. multiVehicleEnabled() returns true and STRATUM does not override QGCOptions — multi-vehicle instantiation is already live . Provides vehicles(), selectedVehicles, and vehicleAdded/vehicleRemoved.
MavCommandQueue	Constructed per Vehicle (Vehicle.cc:266). Retry and ACK matching are scoped to one vehicle's queue. Three concurrent commands to three vehicles are safe by construction ; no new queuing infrastructure is required.
VehicleLinkManager	Per-Vehicle, already many-to-many on links. The NEXAM_TARGET_SELECT broadcast fix (d2d616a) established the correct idiom: enumerate <i>this vehicle's</i> links, never "the" link.
FactGroups	Every group is a per-Vehicle member, including the Nexam engagement and target-track groups and the GNSS horizontalAccuracy / verticalAccuracy facts. Three vehicles yield three independent fact trees with no plumbing.
VehicleMapItem.qml	Already rendered through a MapItemView over the full vehicles model. The map draws every connected vehicle today.
PlanMasterController	startStaticActiveVehicle(Vehicle*) exists and is already used per-vehicle in FlyViewMap.qml:463. Per-vehicle mission upload is proven if routed approaches are ever needed.
MultiVehicleList.qml MultiVehicleSelector.qml	Stock, present, and gated behind enableMultiVehiclePanel. Inert but available.

2.4 What is not

Only the Nexam FlyView QML singletons. Each is instantiated exactly once and bound to QGroundControl.multiVehicleManager.activeVehicle: StandoffController, EngagementController, the abort

and vision overlays, the Set Standoff panel, every custom `GuidedAction*` button, and — visually — the single surveillance circle and target marker in `FlyViewMap.qml:796–873`.

Two components already do it correctly and serve as the internal precedent: `TrackerRoiOverlay.qml` and `TargetTrackingOverlay.qml` both take `vehicle` as a `required` property rather than reaching for the global. The architecture in §4 generalises that precedent rather than inventing a pattern.

2.5 Five findings that dictate the architecture

FINDING F1 — PX4 OWNS THE HOLD-POINT GEOMETRY

On the wire, a three-vehicle ring mission is three single-vehicle standoffs that differ in `param2` alone. This is the cheapest possible extension of a proven contract, and it removes the trajectory generator from STRATUM's requirements entirely.

FINDING F2 — ARRIVAL IS ALREADY OBSERVABLE

`standoff.cpp:139` calls `request_hold_mode()` exactly once when the terminal pose settles. The vehicle therefore announces its own arrival as a `nav_state 9 → 3` transition. No new telemetry, no vehicle-side change, no arrival estimator.

FINDING F3 — THE RUN-IN FLIES AT THE ALTITUDE LATCHED ON ACTIVATION

`standoff.cpp:238` latches `_approach_alt_amsl` from the current global position at `begin_sequence()`, and APPROACH publishes that altitude. Climb each vehicle to its own transit level *before* commit and the run-ins are vertically separated with **zero firmware change**. Conditional on `STDF_SEQ ∈ {1,3}`.

FINDING F4 — CRUISING SPEED IS TRIMMABLE, BUT ONLY AFTER ACTIVATION

`on_activation` calls `reset_cruising_speed()`, and APPROACH republishes its setpoint every cycle carrying `cruising_speed = get_cruising_speed()` (`standoff.cpp:388`). A `DO_CHANGE_SPEED` sent *after* `nav_state == 9` is confirmed takes effect within one navigator cycle. Sent before, it is discarded.

FINDING F5 — FAILSAFE DEFAULTS ARE HOSTILE TO A RING

No custom mode has bespoke failsafe treatment; all four are bucketed with stock AUTO modes under the generic `AutoModes` exception bit (`failsafe.cpp:504–542`). With stock parameters: `NAV_DLL_ACT = 0` — datalink loss triggers **nothing**; `COM_RCL_EXCEPT = 0` with `NAV_RCL_ACT = 2` — RC loss triggers **RTL**. Three vehicles executing independent, uncoordinated RTLs from three points on a ring is precisely the crossing pattern this subsystem exists to prevent.

3 Architectural decisions

Nine decisions. Each states the option space, the choice, and the consequence the choice imposes. A decision without a stated consequence is a preference, not an architecture.

D1 LOCUS OF COORDINATION

Options. (a) ground-side orchestration in STRATUM; (b) onboard distributed coordination between vehicles; (c) hybrid — ground planning, onboard deconfliction.

Decision: (a). STRATUM plans and supervises; PX4 executes. No inter-vehicle link, no onboard awareness of neighbours.

Rationale. Option (b) requires an inter-vehicle datalink, a shared world model and a distributed-consensus safety case; that is a programme, not a feature, and it is unjustified at $n = 3$. Option (c) requires firmware change to a stack whose own documentation records that it was statically reviewed but never compile-verified in the environment that produced it. Option (a) reuses a proven contract end to end.

Consequence. Ground-side coordination promotes the C2 link from a supervisory element to a mission-critical one. That consequence is paid off by D4: if separation is guaranteed by geometry rather than by supervision, losing the link degrades the mission without endangering the airframes. **The link must not be load-bearing for safety.** This is the single most important structural commitment in this document, and every later decision is checked against it.

D2 COMMAND PRIMITIVE

Options. (a) reuse DO_STANDOFF (31010) + DO_SET_MODE per vehicle; (b) upload per-vehicle missions through MissionManager; (c) stream offboard setpoints from STRATUM.

Decision: (a), with (b) held in reserve for multi-leg routing.

Option (c) is rejected outright: streaming setpoints from a ground station makes the radio link an inner control loop, which fails D1's payoff condition immediately and converts every link dropout into a control-loop failure. Option (b) is over-engineering for a straight-line run-in, but the machinery exists, is already per-vehicle, and remains available if terrain or airspace ever forces routed approaches.

D3 GEOMETRY BOUNDARY

Decision: STRATUM owns *ring geometry and scheduling*. PX4 owns *hold-point geometry and approach staging*. The boundary is the existing 31010 contract, unchanged.

STRATUM computes the ring from (target, R , H); the per-slot bearing θ_i ; the per-vehicle transit level; and the commit schedule. It sends (target, R , θ_i , H) to vehicle i . Nothing else crosses the boundary.

Consequence. The wire contract is untouched, the firmware is untouched, and three vehicles differ by one float. This decision is what keeps the change small enough to verify.

D4 TRANSIT DECONFLICTION MECHANISM

Options. (a) closed-loop ground-commanded avoidance; (b) altitude stratification during run-in; (c) strict temporal sequencing, one vehicle at a time; (d) crossing-free slot assignment.

Decision: (b) + (d) as the primary mechanism, (c) as an operator-selectable conservative mode, (a) rejected.

Why (a) is rejected, precisely. The loop would enforce a metre-class separation criterion using positions known to roughly $\pm 1\text{--}2.5$ m per vehicle, sampled at telemetry rate, transported over a link with variable latency, actuated by a mode change with an ACK round trip. The loop delay is comparable to the time a 15 m/s vehicle needs to traverse the entire keepout volume. Such a controller does not add safety; it adds a failure mode that activates precisely when the link is degraded — that is, when the vehicles are already least supervised. **STRATUM will not command avoidance manoeuvres.**

Options (b) and (d) are free, deterministic, and verifiable before the vehicles leave the ground. §7 develops them.

D5 SLOT ASSIGNMENT POLICY

Decision: STRATUM proposes; the operator disposes. The proposal is the cyclic-order-preserving assignment that minimises total run-in path length. Any override is permitted; an override that induces a path crossing is flagged, not blocked.

"Optimum" was undefined in the original objectives. This decision defines it as *minimum total path length subject to a no-crossing constraint* — the objective that also happens to buy deconfliction. At $n = 3$ the assignment problem is six permutations; enumeration is exact and costs nothing, so no solver is introduced.

The no-crossing constraint has a clean geometric basis: if the cyclic angular order of the vehicles' launch positions about the target matches the cyclic order of their assigned slot bearings, straight-line run-ins do not cross.

D6 SEPARATION ENFORCEMENT SEMANTICS

Decision: the separation function is an advisory monitor that may recommend and, on operator authorisation, command HOLD. It never commands a manoeuvre. The 1 m bubble is retained as the hard body keepout but is not the operative criterion.

§7.5 derives the operative criterion as an uncertainty-inflated separation minimum and moves its enforcement from runtime to *slot-placement time* in the wizard. Deconfliction becomes a constraint in the user interface, not a controller in the air. A constraint that has already succeeded on the ground cannot fail in flight.

D7 ARRIVAL SYNCHRONISATION

Decision: v1 achieves time coherence by staggered commit — an open-loop schedule computed from path length and nominal cruise speed, closed by observation of actual arrival skew. v2 adds post-activation DO_CHANGE_SPEED trim per finding F4.

Whether simultaneous arrival is a requirement at all is still open (§11.2). Staggered commit costs nothing and delivers coherence to within the accuracy of the cruise-speed estimate — a few seconds over a 500 m run-in. If a tighter number is operationally required, F4 supplies the mechanism, and the ordering constraint (**speed after mode activation, never before**) is recorded here so that it is not discovered in flight test.

D8 AUTHORITY AND FAILSAFE POSTURE

Decision: an orchestrated mission cannot reach ARMED until STRATUM has read back and verified a required failsafe parameter block on every participating vehicle. Verification failure blocks the mission and names the offending vehicle and parameter.

Per finding F5 the defaults are wrong for this mission. The mechanism — a gate, not a hope — is the decision; the values are ratified in flight test.

Parameter	Required	Why
NAV_DLL_ACT	non-zero (Hold, or RTL with a deconflicted return altitude)	A vehicle that loses STRATUM must not continue indefinitely unsupervised.
NAV_RCL_ACT	Hold, or COM_RCL_EXCEPT bit 1 set	Three independent RTLs from a ring is the crossing case.
COM_DLL_EXCEPT	explicit, not defaulted	Must be a decision, not an accident.
RTL_RETURN_ALT	per-vehicle, stratified	If RTL is retained, return legs need the same vertical discipline as the run-in.
STDF_SEQ	1 or 3 (approach_with_height = false)	Deconfliction-critical. Values 2 and 4 fold the climb into the approach and collapse the vertical separation §7.3 depends on.
MAV_SYS_ID	unique across the fleet	See D9.

The STDF_SEQ row is not a preference. It is a coupling between a parameter documented as a sensor-settling convenience and a property the safety argument depends on. It must be verified, never assumed.

D9 VEHICLE IDENTITY PROVISIONING

Decision: MAV_SYS_ID is a per-airframe configuration item under production control. STRATUM detects a colliding fleet and refuses the mission.

MAV_SYS_ID defaults to 1 and is unmanaged in the fork; nothing auto-assigns it. MultiVehicleManager dedupes by vehicle id, so two Daggers shipped as delivered will present as **one** vehicle — the second simply never appears. This is a silent failure that looks like a radio problem and is not one. It belongs in the production build and acceptance-test procedure, not in software.

3.10 The architecture in one picture

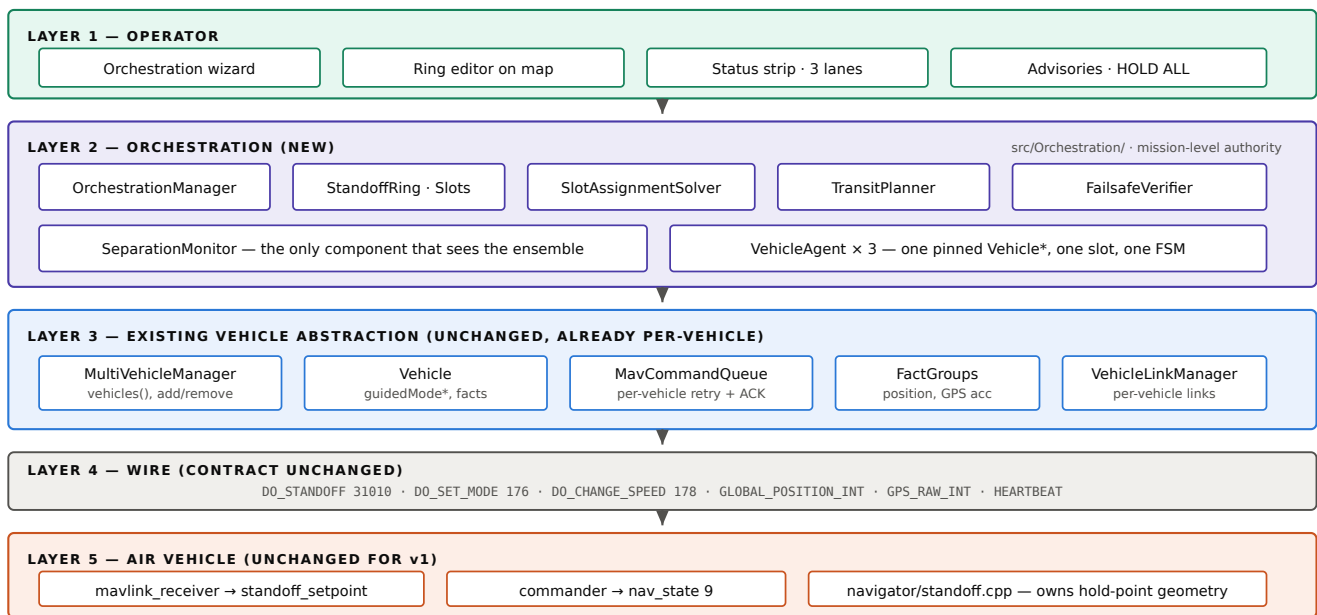


Figure 4 — Layered target architecture. The new subsystem occupies one layer. Layer 3 is reused verbatim; layers 4 and 5 are untouched for v1. The architectural value of that separation is that the existing single-vehicle capability — which works today — cannot regress as a side effect of building the new one.

4 Entity model and component architecture

4.1 Entity model

Two properties of this model carry weight. First, the standoff ring is **derived** and never stored, so it cannot drift out of agreement with the target. Second, `SeparationMonitor` is the only entity with visibility across agents — separation is an emergent property of the ensemble, and only the component that observes the ensemble may make a separation claim. Nothing else in the system is permitted to assert one.

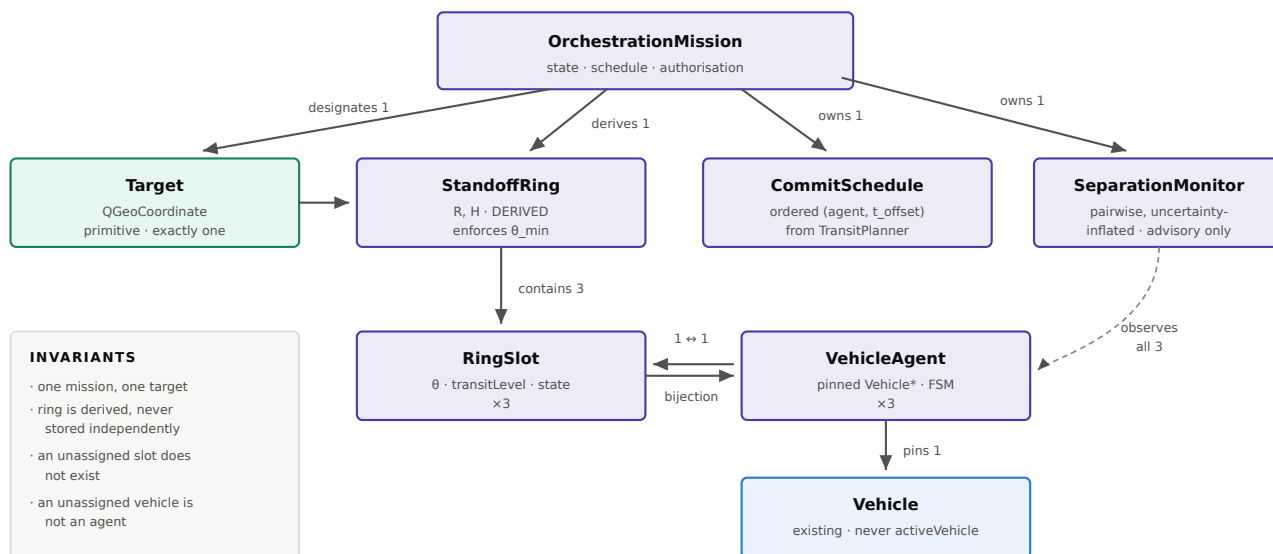


Figure 5 — Entity-relationship model. Cardinalities are enforced, not conventional: the slot-agent relation is a bijection, so a partially assigned ring is not a representable state. Multi-target operation is out of scope because it is a different architecture, not a larger one.

4.2 Component decomposition

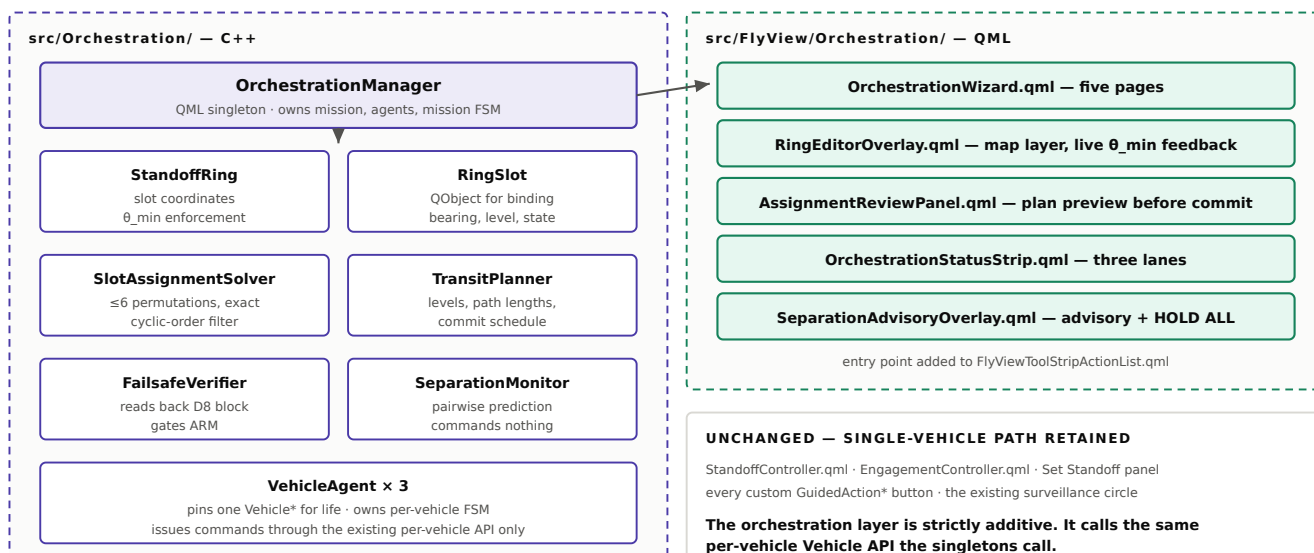


Figure 6 — Orchestration module decomposition. Registration follows the fork's established pattern: `add_subdirectory(Orchestration)` plus a `target_link_libraries` entry in `src/CMakeLists.txt`, and a `Q_PROPERTY` on `QGroundControlQmlGlobal` alongside `multiVehicleManager`.

5 Data flow architecture

5.1 Command path

The command path for one agent is shown in full in Figure 7. Two properties are worth stating before reading it. Every step is gated on an acknowledgement or an observed state, so the agent never assumes a command landed. And every element of the path below the agent already exists and is already per-vehicle, so three agents running this path concurrently contend for nothing.

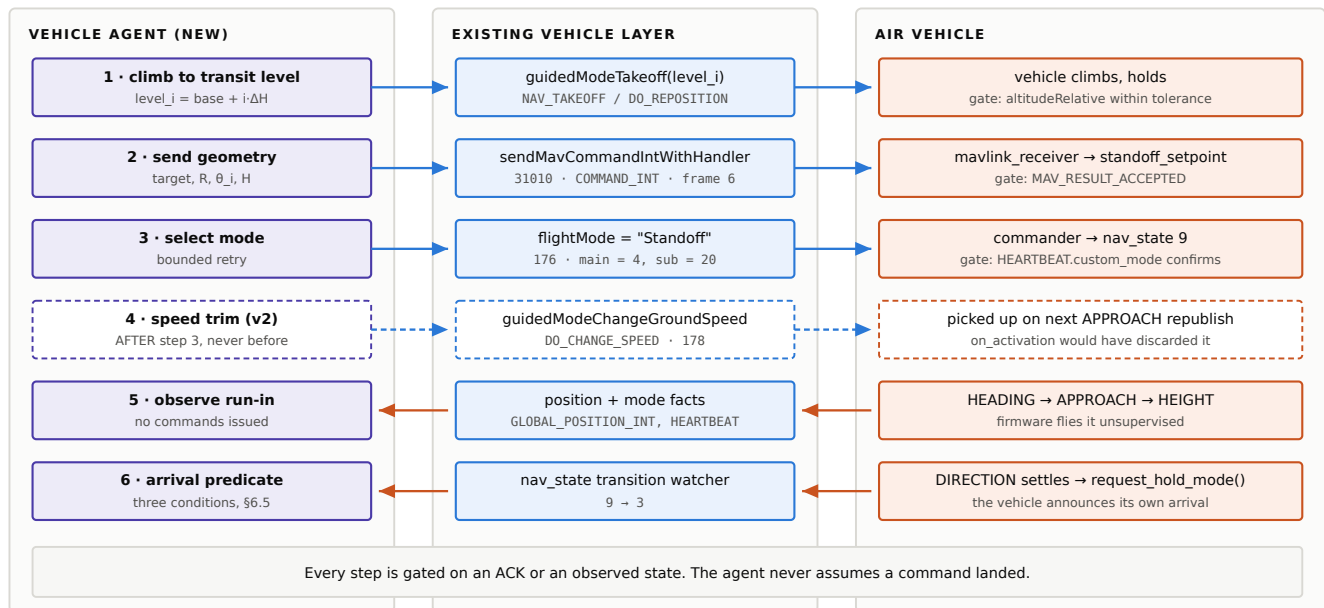


Figure 7 — Command data flow, one agent end to end. Steps 1–4 are commands issued down the stack; steps 5–6 are observations flowing back up. Step 4 is dashed because it is deferred to v2 — and its ordering constraint, derived from finding F4, is recorded on the diagram so it cannot be lost.

5.2 Telemetry path and the separation monitor

The monitor consumes three streams per vehicle and produces one advisory. It is the only component permitted to reason across vehicles, and it is deliberately given no actuation authority — per D6, its output is information for the operator, not a command for an airframe.

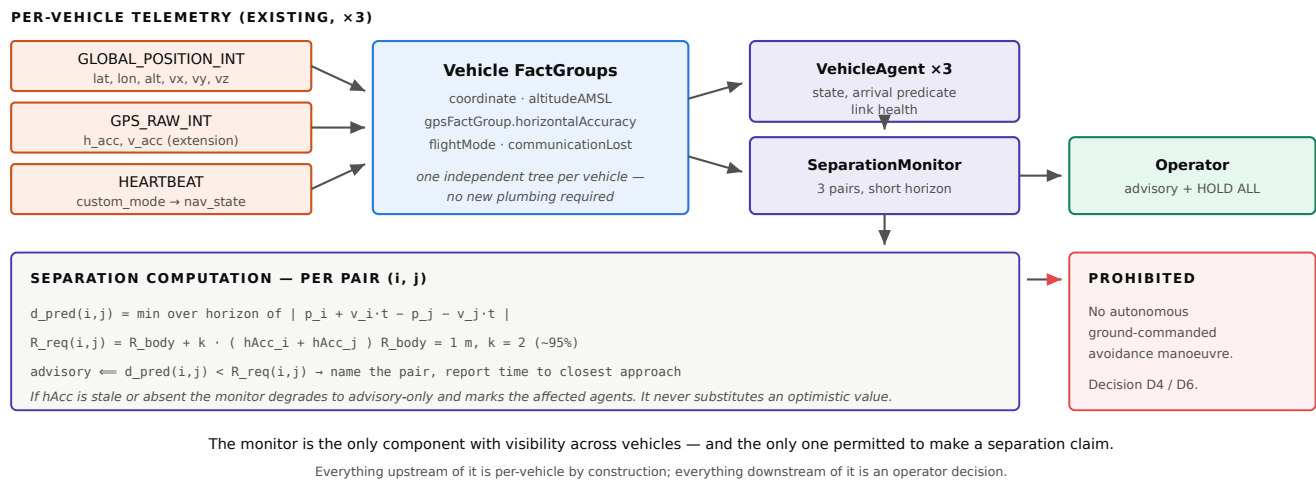


Figure 8 — Telemetry data flow and the separation monitor. The GNSS accuracy facts the monitor needs already exist in STRATUM, added by commit `6df2bc3` for a display purpose. They are re-purposed here as the input that makes the separation criterion honest.

5.3 Wire contract — unchanged

No message, field or identifier changes. The full contract is reproduced in Appendix A for reference during implementation; the summary is that a three-vehicle mission uses the same six message types a one-vehicle mission uses today, addressed to three system identifiers instead of one.

6 Behavioural architecture

6.1 Operator workflow

The wizard exists to make the plan reviewable before it is irreversible. Everything up to page 5 is editable and costs nothing; page 5 puts vehicles in the air. The constraint that guarantees terminal separation is enforced on page 3, on the ground, where failing it is an inconvenience rather than an incident.

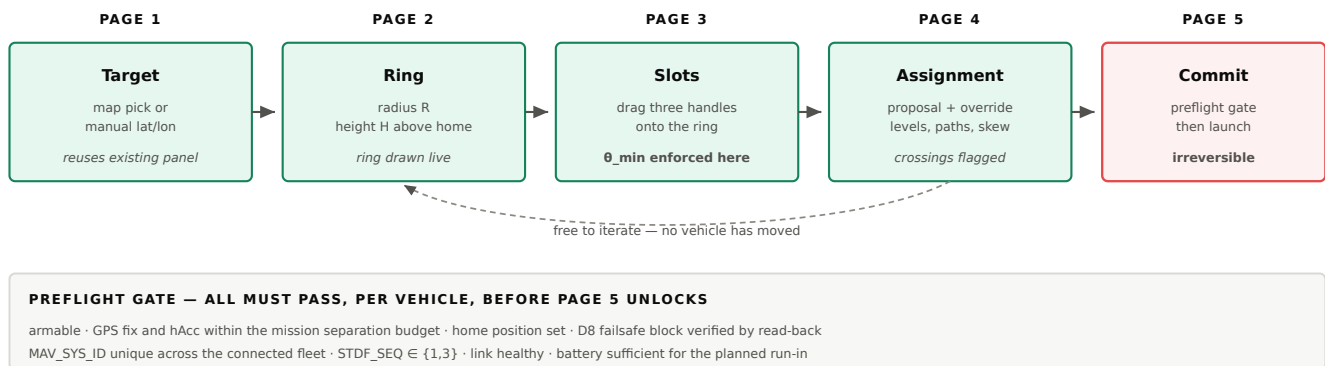


Figure 9 — Operator workflow. Five pages, one irreversible step. The preflight gate is where decision D8 acquires teeth: a mission whose failsafe parameters have not been read back and verified cannot reach page 5.

6.2 Mission state machine

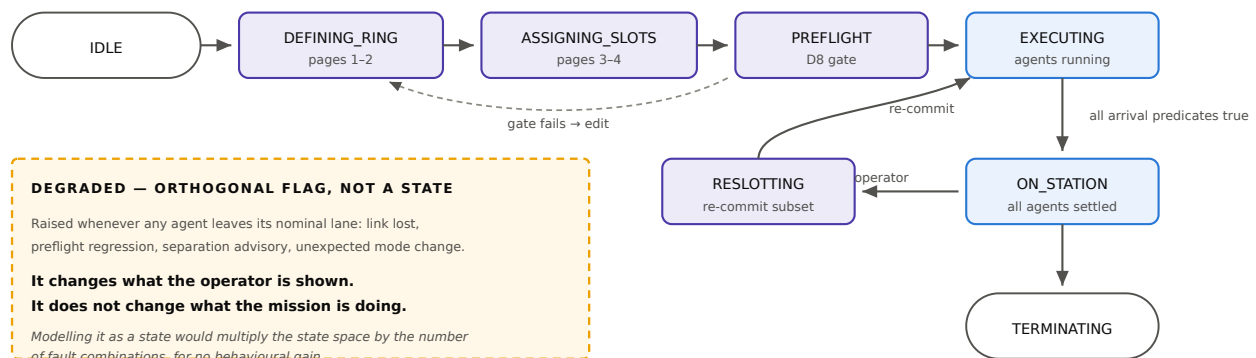


Figure 10 — Mission state machine. Seven states and one orthogonal flag. The mission is editable and reversible up to PREFLIGHT; from EXECUTING onward the operator's controls are hold, re-slot and terminate.

6.3 Agent state machine

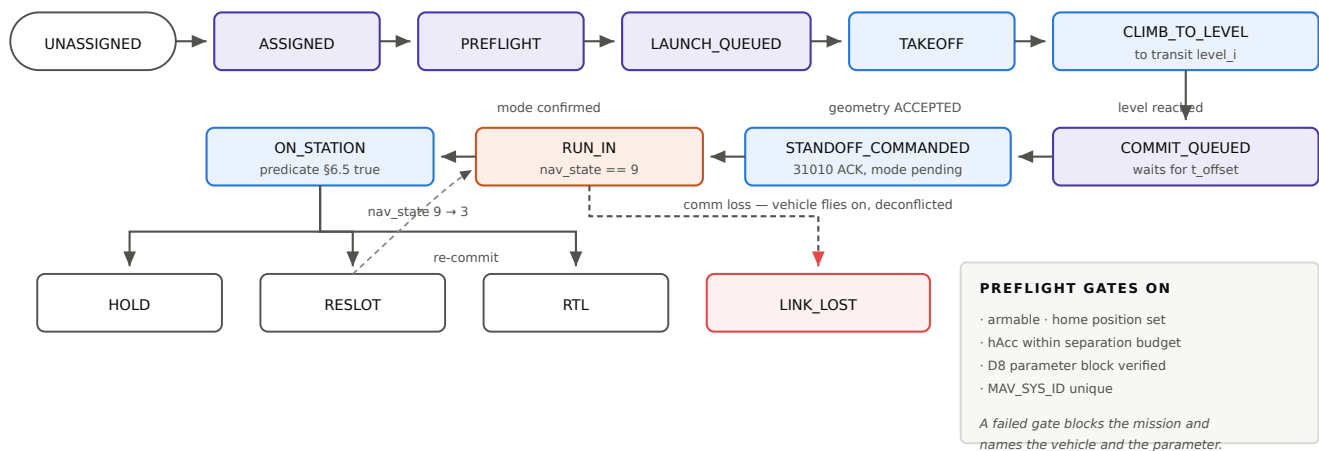


Figure 11 — Agent state machine. An agent pins its vehicle at ASSIGNED and never releases it. Note that LINK_LOST is reachable from RUN_IN without becoming a hazard: the vehicle completes a run-in that was deconflicted before commit. That is decision D1's payoff being collected.

6.4 Commit schedule and arrival coherence

Commit is staggered so that arrivals converge. STRATUM estimates each run-in duration from path length and nominal cruise speed, then delays each agent's commit by the difference. The schedule is open-loop; the skew is measured and reported, and that measurement is what informs whether the v2 speed trim is worth building.

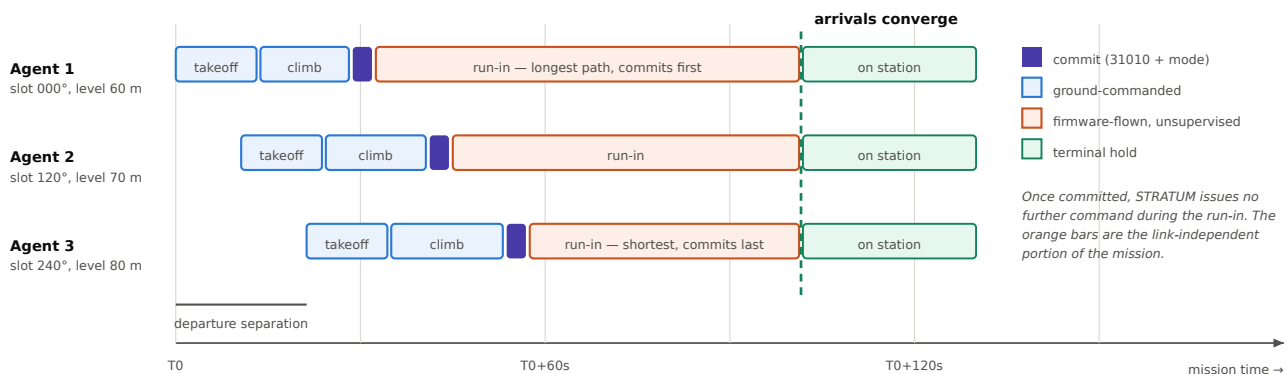


Figure 12 — Commit schedule, three agents. Departures are separated in time; commits are staggered inversely to path length so that arrivals converge. The orange spans are the periods during which the mission does not depend on the C2 link at all — the practical meaning of decision D1's payoff condition.

6.5 Arrival predicate

The $9 \rightarrow 3$ transition alone is not sufficient, because a vehicle can reach Hold by other routes — operator action, or a failsafe. ON_STATION therefore requires all three of the following, at negligible cost:

1. `nav_state` transitioned $9 \rightarrow 3$ (the firmware's own arrival announcement)
2. horizontal distance to the computed slot coordinate $< \text{NAV_ACC_RAD}$
3. $|\text{altitudeAMSL} - (\text{home_alt} + H)| < \text{altitude acceptance radius}$

6.6 Re-slotting

An `ON_STATION` vehicle is in Hold, not Standoff, so re-slotting repeats the full commit sequence with the new bearing. A vehicle still in `RUN_IN` may be re-slotted by sending a fresh 31010 alone — `standoff.cpp:117` restarts the staged manoeuvre from phase 1 on any new setpoint — but that replays the in-place yaw, which reads as abrupt to an operator watching the video feed. Prefer re-slotting from Hold; permit it from run-in.

7 Deconfliction architecture

7.1 Reframing the problem

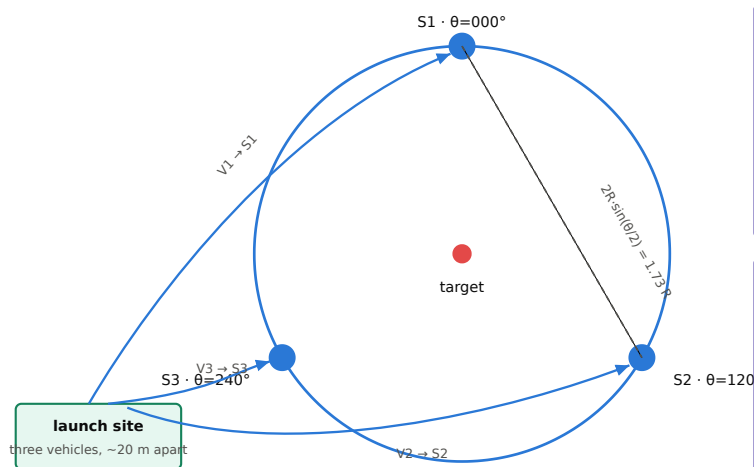
The 1 m bubble was specified as a station-keeping guard. The geometry says it is not needed there. Chord separation between two slots on a ring of radius R separated by angle θ is $2R \cdot \sin(\theta/2)$.

Ring radius R	$\theta = 120^\circ$ (even)	$\theta = 30^\circ$	$\theta = 10^\circ$	$\theta = 1^\circ$
50 m	86.6 m	25.9 m	8.7 m	0.87 m
100 m	173.2 m	51.8 m	17.4 m	1.75 m
300 m	519.6 m	155.3 m	52.3 m	5.24 m

THE CONCLUSION THIS FORCES

The terminal state is deconflicted by ring geometry. For two vehicles to approach the keepout volume on a 300 m ring they would have to be placed less than one degree apart. The conflict lives entirely in the transit phase, and specifically near the launch point, where three vehicles may depart from within tens of metres of one another. Any architecture that spends its effort on station-keeping separation is solving the wrong problem.

PLAN VIEW — CROSSING-FREE ASSIGNMENT PRESERVES CYCLIC ORDER



LAYER 1 — CONSTRAINT AT SLOT-PLACEMENT TIME

$$\theta_{\min} = 2 \cdot \arcsin(R_{\text{req}} / (2R))$$

Slot handles cannot be dragged closer than $\theta_{\min}$.
Terminal separation therefore becomes a UI constraint evaluated before flight, not a controller in the air.

A constraint that has already succeeded on the ground cannot fail in flight.

LAYER 3 — CYCLIC-ORDER ASSIGNMENT

If the cyclic angular order of launch positions about the target matches the cyclic order of assigned slot bearings, straight-line run-ins do not cross.

At $n = 3$ this is six permutations — enumerated exactly, scored by total path length, filtered by the constraint.

No solver is introduced. None is needed.

Figure 13 — Deconfliction, plan view. Ring geometry separates the terminal stations by hundreds of metres; the run-in paths are kept from crossing by preserving cyclic order. Neither mechanism runs in flight — both are settled before the first vehicle lifts.

7.2 Layer 1 — structural, at slot-placement time

The wizard enforces $\theta_{\min} = 2 \cdot \arcsin(R_{\text{req}} / (2R))$ as a drag constraint on page 3. Figure 16 shows how weak that constraint is in practice at any operationally sensible ring radius — which is the point. The constraint costs the operator nothing and removes an entire failure class.

7.3 Layer 2 — vertical stratification during transit

This is the workhorse, and it costs one extra climb command per vehicle. Per finding F3, the APPROACH phase flies at the altitude latched when Standoff activates. Climb each vehicle to its own level before commit and the run-ins are vertically

separated whatever their lateral paths do. The HEIGHT phase then brings each vehicle to the common hold altitude only once it is over its own slot, by which point Layer 1 guarantees the lateral separation.

ELEVATION — TRANSIT LEVELS ARE DISTINCT; HOLD HEIGHT IS COMMON

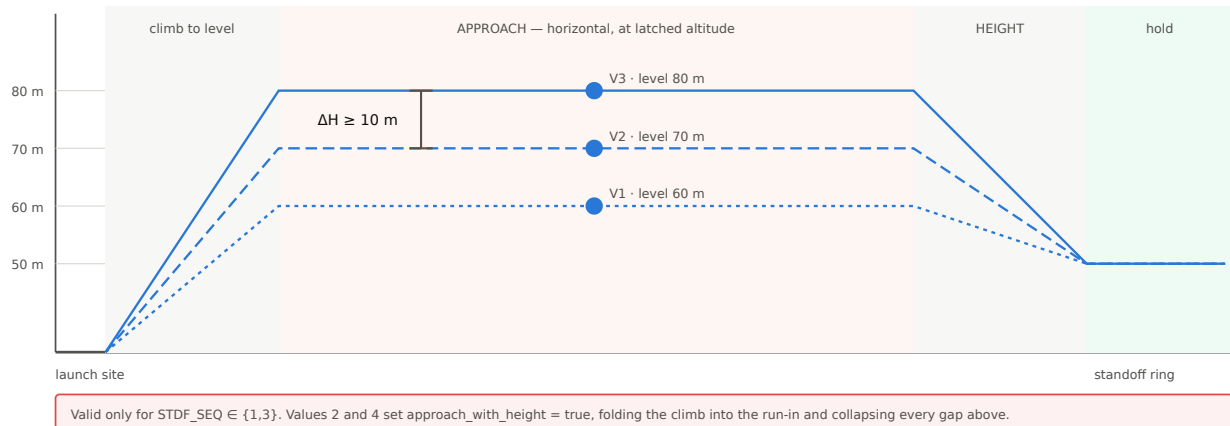


Figure 14 — Deconfliction, elevation. Lateral path crossings during the APPROACH phase are irrelevant because the paths are vertically separated. Vehicles converge to the common hold height only after they are over their own slots, where Layer 1 has already guaranteed hundreds of metres of lateral separation. $\Delta H = 10$ m is chosen against a metre-class vertical position budget with wide margin.

7.4 Layer 3 — assignment and schedule

Cyclic-order-preserving assignment (D5) removes lateral crossings for straight-line run-ins; staggered commit (D7) separates departures in time. Neither costs anything, and both are computed in the planner before commit.

7.5 Layer 4 — the monitor, and what the bubble actually is

The operative separation minimum is not a constant. It is a function of what the system actually knows about where the vehicles are:

```
R_req(i,j) = R_body + k · ( hAcc_i + hAcc_j )

R_body = 1 m      hard body keepout – the nominal "phantom bubble"
hAcc   = reported GNSS horizontal accuracy, per vehicle
        (already in STRATUM: vehicle.gpsFactGroup.horizontalAccuracy, from
        GPS_RAW_INT h_acc – added by commit 6df2bc3 for a display purpose)
k       = 2       for approximately 95% confidence
```

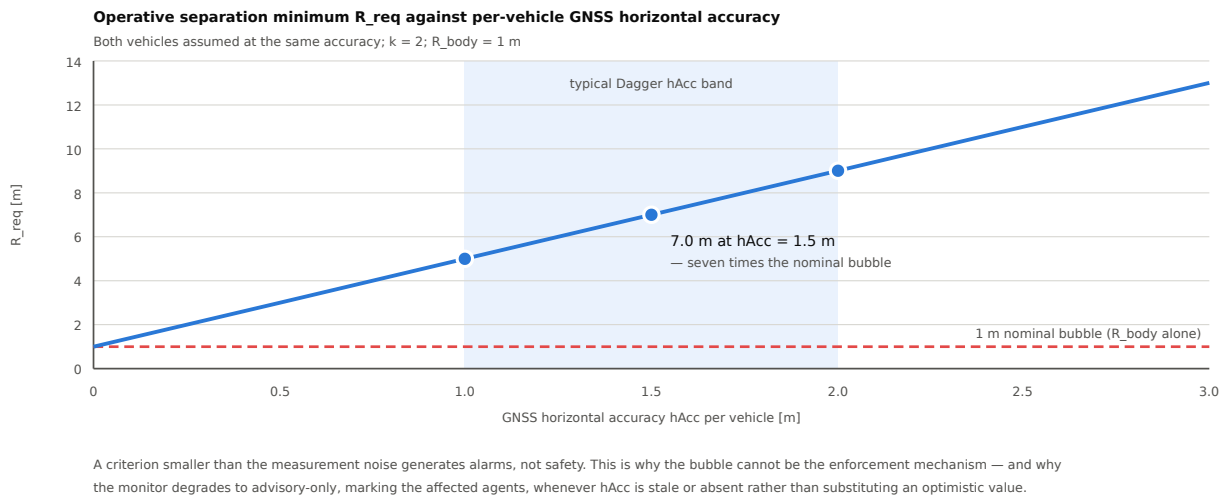


Figure 15 — Operative separation minimum against GNSS accuracy. That the operative minimum is roughly seven times the specified bubble at typical accuracy is not a criticism of the specification. It is the reason the bubble cannot be the enforcement mechanism.

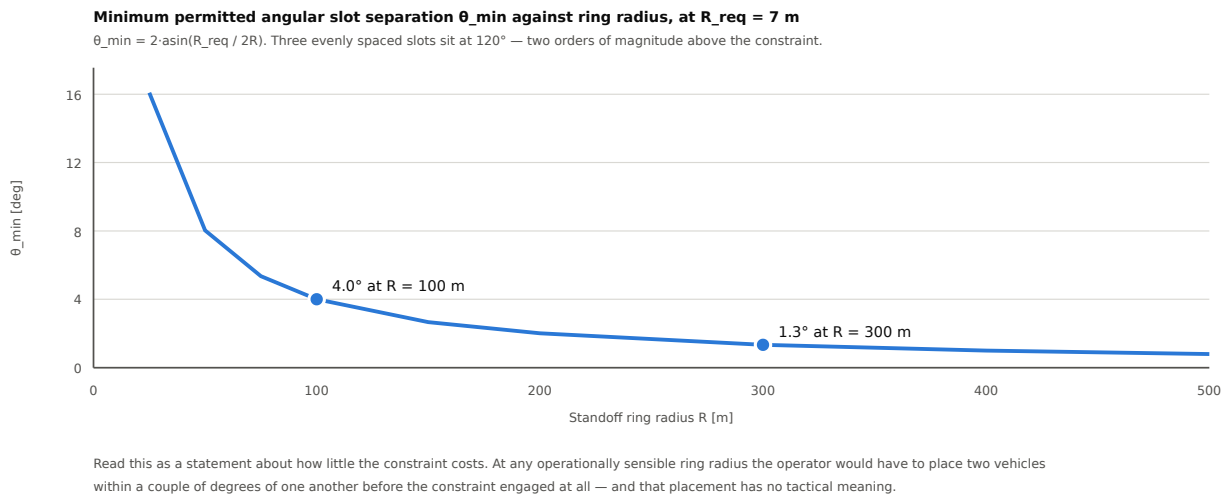


Figure 16 — Minimum angular separation against ring radius. The constraint is almost free. It exists to remove a failure class that no reasonable operator would enter deliberately, and that an unreasonable one could enter accidentally.

8 Failure and degraded modes

A degraded-mode table is the honest test of an architecture, because it is where optimistic assumptions surface. The organising claim below is that the C2 link is not load-bearing for safety after commit — and the table is where that claim is either kept or broken.

FAULT → CONSEQUENCE → RESPONSE, BY MISSION PHASE

	BEFORE COMMIT	RUN-IN (LINK-INDEPENDENT)	ON STATION
Link loss, one vehicle	agent blocked at PREFLIGHT nothing has moved	vehicle completes its run-in and holds on its own slot DEGRADED, not hazardous	holds; NAV_DLL_ACT governs after the configured timeout
Link loss, all vehicles	mission cannot start	all three complete and hold — deconflicted before commit	NAV_DLL_ACT must be explicit — stock default does nothing
One vehicle aborts / RTLs	de-scope to two vehicles and re-plan, or clear fault	remaining agents continue as briefed	ring NOT re-solved automatically STRATUM offers the re-plan
Separation advisory	θ_{min} prevents it — cannot be committed	advisory + HOLD ALL offered no autonomous manoeuvre	geometrically unreachable
GNSS accuracy degrades	preflight gate blocks	R_req grows; a compliant geometry may now advise	correct behaviour — the operator is told why the advisory appeared
The centre column is the architecture's load test: every entry is amber or better without the link. That is decision D1's payoff, collected.			

Figure 17 — Degraded-mode behaviour map. Green = fault contained before anything moves. Amber = mission degraded, airframes safe. Red = requires configuration the stock firmware does not supply, and is therefore closed by decision D8 rather than by hope.

8.1 The one entry that is not self-closing

Battery failsafe is uncoordinated by design: no mode-aware deferral exists in the firmware, and none is proposed. A critical-battery event will act on whatever nav state the vehicle is in, independently of the other two. The architectural response is not to suppress it — suppressing a battery failsafe to preserve a formation is an inversion of priorities — but to make it visible early. STRATUM must display remaining endurance per agent against the planned run-in and hold duration, so that the operator sees it coming rather than discovering it.

9 Firmware deltas and implementation staging

9.1 Firmware deltas required

FOR VERSION 1: NONE

The architecture rides entirely on the existing 31010 contract, the existing Standoff mode, the existing `request_hold_mode()` hand-off, and the existing `STDF_SEQ` staging. Everything that must change is either a STRATUM addition or a parameter value. This is the strongest single argument for the design, and it is the reason the verification burden is tractable.

Non-blocking firmware candidates, recorded here so they are not rediscovered later:

- A sequence or echo identifier on `standoff_setpoint` would let STRATUM confirm *which* setpoint a vehicle is flying, not merely that one was accepted. Useful when re-slotting under a degraded link, where an unacknowledged re-command and a stale one are currently indistinguishable.
- The asymmetry between `EngagementDive` — which, on target loss, levels and waits under the same nav state with no self-abort — and `VisionEngagement`, which self-aborts to `nav_state` 16, should be resolved before any multi-vehicle engagement is contemplated. It is out of scope for v1 and blocking for §11.5.

9.2 Staging

STAGING — TECHNICAL RISK IS CONCENTRATED IN S0-S2

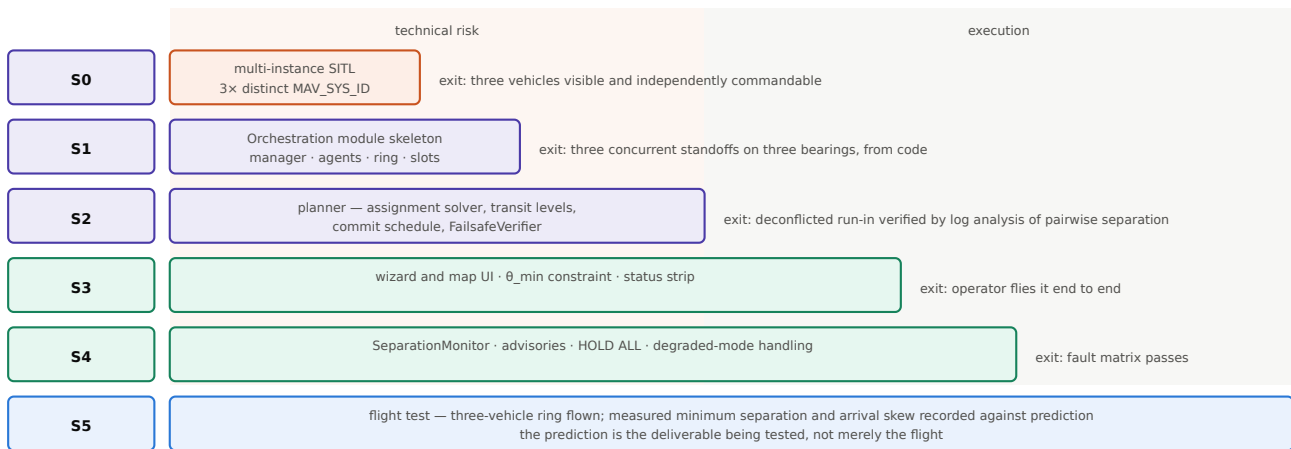


Figure 18 — Implementation staging. S0's exit criterion is deliberately a real multi-instance SITL run rather than a static argument, because the firmware documentation itself records that these custom modes were statically reviewed but never compile-verified in the environment that produced them.

10 Risk register

#	Risk	Class	Mitigation
R1	MAV_SYS_ID collision across shipped airframes — the second vehicle silently never appears, and the symptom resembles a radio fault	Production configuration	D9 — per-airframe configuration item in the build and acceptance procedure; STRATUM detects and refuses
R2	NAV_DLL_ACT = 0 default — datalink loss triggers no action at all	Systemic, latent	D8 — verified parameter block gates ARM
R3	NAV_RCL_ACT = 2 default with no auto-mode exception — three uncoordinated RTLs from a ring	Systemic, emergent	D8 — Hold, or exception bit set; stratified RTL_RETURN_ALT if RTL is retained
R4	STDF_SEQ $\in \{2, 4\}$ silently collapses the transit stratification the safety argument depends on	Design coupling	D8 — STDF_SEQ declared deconfliction-critical and verified, never assumed
R5	Custom MAVLink identifiers 42001–42006 are a project-local allocation	Interoperability	Accepted for now and documented; revisit before any mixed-dialect fleet
R6	The custom firmware modes were statically reviewed but, by their own documentation, never compile-verified in the environment that produced them	Process	S0 exit criterion is a real multi-instance SITL run
R7	EngagementDive has no self-abort on target loss, unlike VisionEngagement	Safety asymmetry	Out of scope for v1; blocking for the ruling in §11.5
R8	Ground-side orchestration raises C2 link criticality from supervisory to mission-critical	Architectural	D1 + D4 — separation is guaranteed by geometry, so the link is not load-bearing for safety after commit. Figure 17 is the evidence.

11 Open decisions requiring ruling

These are not gaps. They are the points at which the architecture is deliberately parameterised pending an operational answer, and each is stated with the default the design will take if no ruling is given.

§	Question	Default if unruled	What changes with the answer
11.1	Common or per-slot hold height?	Common H , distinct transit levels	The wire contract already carries z per vehicle, so per-slot heights are trivially supported — but they change the sensor geometry and the endurance picture.
11.2	Is arrival skew a requirement with a number?	Best-effort via staggered commit; skew measured and reported	If "all three on station within T seconds" is real, state T and the v2 speed trim moves from deferred into S3.
11.3	Launch topology — single pad or dispersed?	Single pad; departure separation in time as well as altitude	Dispersed launch points make the cyclic-order assignment materially more useful and relax the departure schedule.
11.4	On losing a vehicle mid-mission, does the ring re-solve?	Hold as briefed; offer the re-plan	Automatic re-spacing of two survivors is implementable, but it is a re-commit under degraded conditions and should be an operator decision.
11.5	Does an orchestrated standoff ever transition to engagement?	Excluded from this subsystem's remit	The scoping question that matters most. Both capabilities live in the same application. Multi-vehicle coordination plus terminal engagement is a materially heavier safety case — a different authorisation chain, a different abort architecture, and a different verification burden, compounded by risk R7. It should be scoped in deliberately or excluded explicitly. It should not arrive by accident.

A Appendix — wire contract and parameter reference

A.1 Standoff command — DO_STANDOFF

Unchanged from the single-vehicle contract. Sent once per vehicle; only `param2` differs between the three.

Field	Value	Notes
command	31010	MAV_CMD_USER_1, repurposed as DO_STANDOFF
message	COMMAND_INT	Preserves 1e7 lat/lon precision. <code>COMMAND_LONG</code> works but loses it.
frame	MAV_FRAME_GLOBAL_RELATIVE_ALT (6)	Must be a global frame. A local frame scales x/y by 1e4 and corrupts the target.
x	target latitude × 1e7	The target , not the hold point
y	target longitude × 1e7	—
z	H — height above home [m]	Relative , not AMSL; PX4 adds home altitude
param1	R — standoff distance [m]	Common to all three vehicles
param2	θ_i — bearing [deg, $\theta = N, CW$]	The only field that differs per vehicle
param3	0	—
param4	NaN	Let PX4 choose final yaw; it faces the target
ack	ACCEPTED / DENIED	Immediate; DENIED when lat/lon are not finite. Addressed to the originating sysid/compid.

A.2 Mode selection and supporting commands

Command	Parameters	Use
DO_SET_MODE (176)	base_mode CUSTOM_MODE_ENABLED main = 4 (AUTO), sub = 20	Enter Standoff. Sub-mode value must be 20.
DO_CHANGE_SPEED (178)	groundspeed [m/s]	v2 arrival trim. Send only after nav_state 9 is confirmed — on_activation calls <code>reset_cruising_speed()</code> .
NAV_TAKEOFF / DO_REPOSITION	transit level	Climb to the per-vehicle transit level before commit.
HEARTBEAT	custom_mode = (sub<<24) (main<<16)	Mode read-back; the source of the 9 → 3 arrival transition.
GLOBAL_POSITION_INT	lat, lon, alt, vx, vy, vz	Position and velocity for the separation monitor.
GPS_RAW_INT	h_acc, v_acc (extension fields)	Accuracy input to <code>R_req</code> . Zero is treated as "not provided", not as zero error.

A.3 Nav states and sub-modes in the Dagger firmware

Mode	nav_state	custom sub-mode	Slot origin
Standoff	9	20	reused FREE3
Engagement	11	21	reused unused slot
Abort	16	22	reused FREE1
Vision Engagement	31	23	appended
PN Engagement	–	24	appended
Hold (arrival target)	3	3	stock AUTO_LOITER

A.4 Parameters referenced by this architecture

Parameter	Default	Role in this architecture
STDF_SEQ	1	Deconfliction-critical. Must be 1 or 3. Values 2 and 4 set approach_with_height = true.
NAV_DLL_ACT	0 (disabled)	Must be made explicit and non-zero — D8.
NAV_RCL_ACT	2 (Return)	Must be Hold, or COM_RCL_EXCEPT bit 1 set — D8.
COM_RCL_EXCEPT	0	Auto-mode exception bit for RC loss.
COM_DLL_EXCEPT	0	Auto-mode exception bit for datalink loss.
RTL_RETURN_ALT	stock	Stratify per vehicle if RTL is retained as the loss action.
MAV_SYS_ID	1	Must be unique per airframe — D9, risk R1.
NAV_ACC_RAD	stock	APPROACH acceptance; second clause of the arrival predicate.
NAV_MC_ALT_RAD	stock	HEIGHT acceptance; third clause of the arrival predicate.
MIS_YAW_ERR / MIS_YAW_TMT	stock	HEADING phase acceptance and its timeout bound.
MPC_YAW_MODE	–	Set to 3 (towards waypoint) to keep the nose on target throughout the approach.

A.5 Source references

Location	What it establishes
STRATUM · FlyViewWidgetLayer.qml:266–444	Set Standoff panel; the existing operator entry point
STRATUM · StandoffController.qml	Single-instance controller bound to <code>activeVehicle</code>
STRATUM · Vehicle.cc:1927	<code>guidedModeStandoff</code> ; range check against a shared setting
STRATUM · Vehicle.cc:266	<code>MavCommandQueue</code> constructed per vehicle
STRATUM · PX4FirmwarePlugin.cc:447	31010 emission with the parameter mapping in A.1
STRATUM · FlyViewMap.qml:427–436, 463–479	Map already iterates all vehicles; per-vehicle <code>PlanMasterController</code>
STRATUM · MultiVehicleManager.cc:99	Multi-vehicle gate; <code>multiVehicleEnabled()</code> not overridden
STRATUM · commit 6df2bc3	GNSS <code>horizontalAccuracy</code> / <code>verticalAccuracy</code> facts
STRATUM · commit d2d616a	Per-vehicle link enumeration idiom
PX4-NX · mavlink_receiver.cpp:536, 832	31010 constant; decode, publish and ack
PX4-NX · navigator/standoff.cpp:117, 139, 238, 388	Re-command restart; hold hand-off; latched approach altitude; cruising speed
PX4-NX · commander/failsafe/failsafe.cpp:504–542	Custom modes bucketed under the generic <code>AutoModes</code> exception
PX4-NX · docs_standoff_mode.md	Design rationale, phase staging, verification status

B Closing argument

Because PX4 owns the standoff geometry, STRATUM does not need a trajectory generator; it needs a scheduler and a constraint. Because the firmware latches the run-in altitude at mode activation, vertical deconfliction is available for the price of one climb command. And because the ring geometry separates the terminal stations by hundreds of metres, the 1 m bubble was guarding the one phase of flight that was never in danger.

The scan changed three things about this design, and each change made it smaller. That is usually the sign of an architecture converging on the grain of the system rather than fighting it. The remaining decisions in §3 are then largely about restraint: not streaming setpoints, not commanding avoidance, not re-solving the ring under fire, not refactoring a working single-vehicle path to accommodate a new one.

The residual risk is not in the flight geometry. It is in the failsafe parameter block, where the stock defaults would have three vehicles making independent, uncoordinated decisions at exactly the moment the ground station stops watching. That is why D8 is a gate rather than a recommendation, and why the preflight check reads back parameters instead of trusting them.

Decisions dictate destiny.

NXM-SW-ARCH-STRATUM-MVO-001 v0.2 · Nexam Systems · Company Proprietary · 03 September 2026

Derived from a direct scan of STRATUM-Desktop @ 4095ba3 and PX4-Autopilot-NX. Decisions D1–D9 proposed for review; rulings 11.1–11.5 outstanding.